Some observations about Łukasiewicz algebras

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1 Introduction

These notes are a write-up of some experiments I did using Python programs to study some finite-valued Łukasiewicz logics. In the next section, I sketch some background information, most of which is based on discussion in Urquhart [5] and in Malinowski [3]. My first encounter with these systems, however, was many years ago[1] in a discussion of truth tables in Chapter VII of my now very beat up copy of Lewis and Langford [2]. In particular, I define what I mean by a Łukasiewicz algebra (or Ł-algebra).

In the next several sections, I attempt to summarize and share some of my observations. I suspect that most of the results are already known to specialists.[2]

In the section on divisor lattices, I attempt to make some sense of the observations by showing a connection between Ł-algebras and certain lattices of divisors. This connection opens up a separate connection between certain modal logics and the Ł-algebras.

2 Background

In the 1930s, Polish logician Jan Łukasiewicz developed a family of multi-valued propositional logics[3] based on logical negation and material implication. In the system, propositions are assigned truth values in the interval $[0, 1]$ where the values 0 and 1 represent falsity and truth and values in-between represent degrees of truthfulness. (One of the problems that Łukasiewicz considered was how to deal with statements involving future events.) My interest here is in mathematical relationships between the various logics and and not in any philosophical implications.

2.1 Rules of Formation

The well-formed formulas (or *wffs*) of each of the systems are characterized by the same three rules:

1. a propositional variable is a wff;
2. if X is a wff, then $\neg X$ is a wff; and
3. if X and Y are wffs, then $(X \supset Y)$ is a wff.

In practice, if the expression is enclosed in a pair of outside parentheses, we typically remove that outermost pair. For example:

- "$p \supset p$" stands for "$(p \supset p)$";
- likewise: "$(p \supset q) \supset ((q \supset r) \supset (p \supset r))$" for "$((p \supset q) \supset ((q \supset r) \supset (p \supset r)))$".

[1] late 1960s!

[2] At present, I am not affiliated with a research university, so I don't have ready access to recent research.

[3] herein referred to as Łukasiewicz algebras or Ł-algebras

The propositional variables are the letters p, q, r, ... We assume a countably infinite supply of propositional variables, though, in practice, we only need a few. The prefix operator C stands for conditional, another name for implication.

We can largely[4] avoid parentheses by using Polish (or prefix) notation. When using prefix notation, the three rules are replaced by the following rules:

1. a propositional variable is a wff;
2. if x is a wff, then Nx is a wff; and
3. if x and y are wffs, then Cxy is a wff.

The same lower case letters are used as propositional variables, but the infix operators $\neg *$ and $(* \supset *)$ are replaced by the prefix operators $N*$ and $C**$.

2.2 Basic semantics

As noted, the variables take on certain truth values, and based on these truth values, the wffs take on particular values. Given a wff X, the truth value of X is denoted $|X|$.

The interpretation of the wff $\neg p$ is "it is false that p" or "it is not the case that p", usually expressed more compactly as "not-p". Our truth values are taken in the interval $[0, 1]$ with 0 being interpreted as false and 1 as true. For $\neg p$:

$$|\neg p| := 1 - |p| \qquad \text{negation} \qquad (2.1)$$

The wff $(p \supset q)$ is interpreted as "if p then q" or "p is a sufficient condition for q". For the Ł-algebras, we define the value as follows:

$$|p \supset q| := \min(1,\ 1 - |p| + |q|) \qquad \text{material implication} \qquad (2.2)$$

For examples, consider truth tables for the operators in the three-valued algebra Ł$_3$ (see table 2.1) and the four-valued algebra Ł$_4$ (see table 2.2).

p	$\neg p$
F	T
I	I
T	F

$p \supset q$	q		
p	F	I	T
F	T	T	T
I	I	T	T
T	F	I	T

F=0
I=1/2
T=1

Table 2.1: Semantics for primitive operators in Ł$_3$

Note that with these semantics, a false proposition implies any proposition, and when the antecedent in an implication is true, the implication takes on the value of the consequent. Note also that the matrix for material implication has diagonal bands that run from the lower right to the upper left. These are more pronounced in the matrix for Ł$_4$.

In the finite-valued algebra Ł$_n$, the truth values are uniformly spaced fractions 0, $\frac{1}{n-1}$, $\frac{2}{n-1}$, ..., $\frac{n-2}{n-1}$, 1.

[4] more on that below

p	$\neg p$
F	T
I	J
J	I
T	F

$p \supset q$	q			
p	F	I	J	T
F	T	T	T	T
I	J	T	T	T
J	I	J	T	T
T	F	I	J	T

F=0
I=1/3
J=2/3
T=1

Table 2.2: Semantics for primitive operators in Ł$_4$

2.3 Basic Definitions

The operators $\neg$ and $\supset$ (or N and C when using prefix notation) are completely adequate for developing the algebras, but it is convenient to define several other operators. The first three are very standard:

$$(p \vee q) := Apq := ((p \supset q) \supset q) \quad \text{disjunction} \quad (2.3)$$

$$(p \wedge q) := Kpq := \neg(\neg p \vee \neg q) \quad \text{conjunction} \quad (2.4)$$

$$(p \equiv q) := Epq := ((p \supset q) \wedge (q \supset p)) \quad \text{material equivalence} \quad (2.5)$$

Disjunction is also called “alternation”, hence the prefix ‘A’ in Polish notation. The ‘K’ prefix comes from German. The semantics are straightforward:

$$|p \vee q| = \max(|p|, |q|) \quad (2.6)$$

$$|p \wedge q| = |\min(|p|, |q|) \quad (2.7)$$

$$(p \equiv q) := Epq := \min(1 - |p| + |q|,\ 1 - |q| + |p|) \quad (2.8)$$

Truth tables have been prepared below for Ł$_3$ (Table 2.3) and Ł$_4$ (Table 2.4). The abbreviations F, I, J, and T are used as in Tables 2.1 and 2.2.

$p \vee q$	q		
p	F	I	T
F	F	I	T
I	I	I	T
T	T	T	T

$p \wedge q$	q		
p	F	I	T
F	F	F	F
I	F	I	I
T	F	I	T

$p \equiv q$	q		
p	F	I	T
F	T	I	F
I	I	T	I
T	F	I	T

Table 2.3: Operator semantics for Ł$_3$

2.4 Axioms for Ł$_\infty$ and Ł$_3$

Finding a suitable set of axioms for the Ł-algebras has apparently been a long process. (See [3] and [5]. Those were the principal references for this section.)

$p \vee q$	q			
p	F	I	J	T
F	F	I	J	T
I	I	I	J	T
J	J	J	J	T
T	T	T	T	T

$p \wedge q$	q			
p	F	I	J	T
F	F	F	F	F
I	F	I	I	I
J	F	I	J	J
T	F	I	J	T

$p \equiv q$	q			
p	F	I	J	T
F	T	J	I	F
I	J	T	J	I
J	I	J	T	J
T	F	I	J	T

Table 2.4: Operator semantics for $Ł_4$

Mordchaj Wajsberg proposed the following set of axioms for $Ł_\infty$ in 1935:

$$\vdash p \supset (q \supset p) \qquad \text{positive paradox} \qquad (2.9)$$

$$\vdash (p \supset q) \supset ((q \supset r) \supset (p \supset r)) \qquad \text{syllogism} \qquad (2.10)$$

$$\vdash (\neg p \supset \neg q) \supset (q \supset p) \qquad \text{contrapositive} \qquad (2.11)$$

$$\vdash (p \vee q) \supset (q \vee p) \qquad \text{commutative} \qquad (2.12)$$

$$\vdash (p \supset q) \vee (q \supset p) \qquad (2.13)$$

The semantics of the turnstile ($\vdash$) is that the wff that follows is valid, *i.e.* true under any legal assignment of the variables.

To derive theorems from these axioms, we need two transformation rules. The first, known as *modus ponens* or *detachment*, is easily stated:

$$\text{If } \vdash (X \supset Y) \text{ and } \vdash X, \text{ then } \vdash Y. \qquad \textit{modus ponens} \qquad (2.14)$$

We also need a substitution rule which allows us to uniformly replace each propositional variable by a wff. Some authors avoid this by using propositional schemas in their presentation instead of propositional variables.

There were several proofs in the 1950s that these five axioms and the two rules are complete, *i.e.* that every tautology in $Ł_\infty$ can be derived as a theorem. In addition, it has been shown that Axiom 2.13 is dependent.

In 1931, Wajsberg found a set of axioms for the three-valued Ł-algebra $Ł_3$. These include the Axioms 2.9, 2.10 and 2.11 above along with detachment and substitution and the following axiom:

$$\vdash (p \supset \neg p) \vee p \qquad \text{axiom for } Ł_3 \qquad (2.15)$$

Axioms 2.13 and 2.15 can be thought of as weak forms of the law of the excluded middle, usually written as (EM) $\vdash p \vee \neg p$. Proposition (EM) is valid in the classical propositional calculus (*i.e.* $Ł_2$) but fails for all $n > 2$ in $Ł_n$.

2.5 Axioms for finite Ł-algebras

Algorithms for generating axioms for each of the finite Ł-algebras can be found in both [5] and [3]. From a programming standpoint, R Grigolia's 1977 algorithm is relatively easy to implement.[3, page 40].

We first define two binary connectives:

$$(p + q) := +pq := (\neg p \supset q) \tag{2.16}$$

$$(p \times q) := \times pq := \neg(p \supset \neg q) \tag{2.17}$$

In the classical propositional calculus (or Ł$_2$), these are equivalent disjunction and conjunction. In other Ł-algebras they are not. Truth tables for Ł$_3$ and Ł$_4$ are supplied as Tables 2.5 and 2.6. Note that in Ł$_3$, these differ from disjunction and conjunction in the (I,J) entries. For Ł$_4$, the differences are in the four central entries. The plus operator is stronger than disjunction, and the times operator is weaker than conjunction.

$p+q$	q		
p	F	I	T
F	F	I	T
I	I	T	T
T	T	T	T

$p \times q$	q		
p	F	I	T
F	F	F	F
I	F	F	I
T	F	I	T

Table 2.5: Extended operator semantics for Ł$_3$

$p+q$	q			
p	F	I	J	T
F	F	I	J	T
I	I	J	T	T
J	J	T	T	T
T	T	T	T	T

$p \times q$	q			
p	F	I	J	T
F	F	F	F	F
I	F	F	F	I
J	F	F	I	J
T	F	I	J	T

Table 2.6: Extended operator semantics for Ł$_4$

Using the plus operator, we define a sequence of multiples:

$1p := p$, $2p := p + p$, $3p = 2p + p$, $4p = 3p + p$, and so on.

In Ł$_n$, we find that:

$$\vdash np \subseteq (n-1)p. \tag{2.18}$$

Here is how it plays out in Ł$_4$:

$1p$	$2p$	$3p$	$4p$	notes
F	F	F	F	all false
I	J	T	T	step upward
J	T	T	T	step upward
T	T	T	T	all true

In Ł$_4$, $|\mathfrak{m}\mathfrak{p}| = \mathbb{F}$ for all $\mathfrak{m}$ when $|\mathfrak{p}| = \mathbb{F}$. If $|\mathfrak{p}| \neq \mathbb{F}$, then, for each step in $\mathfrak{m}$, the value of $\mathfrak{m}\mathfrak{p}$ steps up one level until $\mathbb{T}$ is reached. Notice that, from $3\mathfrak{p}$ on, the columns are identical.

This same behavior holds for all finite $\mathfrak{n}$. With larger $\mathfrak{n}$, there are simply more steps.

If this axiom was all that was needed, then the Ł-algebras would form a nested sequence. But they are not nested in a single sequence. Ł$_4$ has theorems that are not theorems in Ł$_3$. There is, however, some nesting. The relationships are just a a bit more complicated. It is however true, that if $\mathfrak{m} < \mathfrak{n}$, then Ł$_\mathfrak{m}$ has theorems that are not theorems of $/_\mathfrak{n}$, in particular:

$$\vdash \mathfrak{m}\mathfrak{p} \subseteq (\mathfrak{m}-1)\mathfrak{p}.$$

The missing axioms are built using the times operator. Instead of coefficients, we have exponents:

$\mathfrak{p}^1 := \mathfrak{p}$, $\mathfrak{p}^2 := \mathfrak{p} \times \mathfrak{p}$, $\mathfrak{p}^3 = \mathfrak{p}^2 \times \mathfrak{p}$, $\mathfrak{p}^4 = \mathfrak{p}^3 \times \mathfrak{p}$, and so on.

In Ł$_4$: Here is how it plays out in Ł$_4$:

$\mathfrak{p}^1$	$\mathfrak{p}^2$	$\mathfrak{p}^3$	$\mathfrak{p}^4$	notes
F	F	F	F	all false
I	F	F	F	step down
J	I	F	F	step down
T	T	T	T	all true

It's basically the opposite of the coefficients. To get axioms from this, we use the following (arcane) schema:

$$\vdash (\mathfrak{n}-1)((\neg\mathfrak{p})^2) + (\mathfrak{p} \times (\mathfrak{j}-1)\mathfrak{p})) \quad \mathfrak{j} \in [2, \mathfrak{n}-2],\ \mathfrak{j} \text{ does not divide } \mathfrak{n}-1 \qquad (2.19)$$

For $\mathfrak{n} = 2$ and $\mathfrak{n} = 3$, there are no axioms of this type. For $\mathfrak{n} = 4$, note that 3 is not divisible by 2, so we get one axiom:

$$3((\neg\mathfrak{p})^2 + (\mathfrak{p} \times 1\mathfrak{p}))$$

Since $1\mathfrak{p} = \mathfrak{p}$ we can simplify this:

$$3((\neg\mathfrak{p})^2 + \mathfrak{p}^2)$$

Although it is valid in Ł$_4$, it fails in Ł$_3$. In Ł$_4$:

		A	B	
$\mathfrak{p}$	$\neg\mathfrak{p}$	$\mathfrak{p}^2$	$(\neg\mathfrak{p})^2$	$A + B$
F	T	F	T	I
I	J	F	I	I
J	I	I	F	I
T	F	T	F	T

In each row: $|A + B| \neq F$, so $|\exists(A + B)| = T$. In Ł$_3$, we have a problem when $|p| = T$:

		A	B	
p	$\neg p$	p^2	$(\neg p)^2$	$A + B$
I	I	F	F	F

In Ł$_3$, when $|p| = I$, we have $|A + B| = F$, so $|\exists(A + B)| = F$.

To conclude, our additional axiom for Ł$_4$ is not valid in Ł$_3$.[5]

2.6 Polish notation, coefficients, and exponents

Polish notation can handle integer coefficients and exponents. The most elegant approach is to define two prefix operators, *e.g.* an underscore for coefficients and a carat for exponents. When we encounter a decimal digit, we require that the preceding characters must also be decimal digits until the prefix operator is reached.

A less elegant solution, the one (unfortunately) adopted in programming here, is to delimit both ends of the decimal number.

If we were writing a Polish notation calculator, and we wanted to add 15 and 20, we could do something like:

+15,20

The comma is a prefix operator that separates the entry of 20 and 15. Processing this from right to left:

```
0: push 0
2: pop -> 0
   push 20     (2x10+0)
,:             (separator)
5: push 5
1: pop -> 5
   push 15     (1x10+5)
+: pop -> 15
   pop -> 20
   push 35     (20+15)
```

From a performance standpoint, use of delimiters in begin-end pairs doesn't make a significant difference.

2.7 Functional incompleteness

With the exception of the classical case (Ł$_2$), none of the systems are functionally complete. The algebras use two primitive connectives, namely $\neg$ and $\supset$ (or

[5] Exercise: Prove or disprove that this axiom will fail in Ł$_n$ whenever n is an odd integer greater than 1.

N and C). Using just these two connectives and composition, it is not possible to define an operator which returns a value other than F=0 or T=1 when the associated value other than F or T. For example, it is impossible to define the either of the unary operators $\iota(\mathrm{p})$ and $\rho(\mathrm{p})$ in Ł$_3$ which have the following truth tables:

p	F	I	T
$\iota(\mathrm{p})$	I	I	I
$\rho(\mathrm{p})$	I	T	F

(The operator ι is a projection and the operator ρ is a rotation.)

If the operator a rotation operator like ρ or ρ^{-1} is added to a finite Ł-algebra, then the resulting system becomes functionally complete. In Rosenbloom [4, *p*52 *ff*], the resulting algebra is referred to as a Post algebra. Hendry in [1] states that adding the operator ι to Ł$_3$ makes that particular system functionally complete.

Whether functional completeness is desirable is a philosophical question that is well beyond the scope of these notes.

3 Results for Ł$_3$

3.1 Definitions

The Python programs went through a number of checks to insure that the various operator were implemented correctly. For the most part, operators were implemented independently of their formal definitions. For example, disjunction and conjunction were implemented as max and min, respectively:

$$|\mathrm{p} \vee \mathrm{q}| := \max(|\mathrm{p}|, |\mathrm{q}|) \tag{3.1}$$

$$|\mathrm{p} \wedge \mathrm{q}| := \min(|\mathrm{p}|, |\mathrm{q}|) \tag{3.2}$$

These are the actual formal semantics. The plus ($+$) and times ($\times$) operators, however, were implemented as defined. To check that the semantics actually worked correctly, the definitions were checked programmatically by substituting all possible combinations of the truth values. The definitions all passed the tests:

```
Target Engine: Łukasiewicz-algebra Ł(3)
        Class: GrigoliaExtensions
```

The target engine says that the truth values are 0, 1/2, and 1 and that negation, material implication, disjunction, conjunction, and material equivalence are set up semantically for Ł$_3$. The class says, in addition, that plus, times, coefficients, and exponents are defined. These are not actual extensions as they can be defined using only negation and material implication.

```
Definitions:
    D1: (p∧q) := ~(~p∨~q)                    conjunction
    D2: (p∨q) := ((p⊃q)⊃q)                   disjunction
    D3: (p≡q) := ((p⊃q)∧(q⊃p))               equivalence
```

These are the actual definitions. The Python program makes sure that both sides agree regardless of the truth value assignments.

```
    D4: (p+q) := (~p⊃q)                      plus
    D5: (p•q) := ~(p⊃~q)                     times
    D6: (2p) := (p+p)                        multiple 2
    D7: (3p) := ((2p)+p)                     multiple 3
    D8: (4p) := ((3p)+p)                     multiple 4
    D9: (p^2) := (p•p)                       exponent 2
   D10: (p^3) := ((p^2)•p)                   exponent 3
   D11: (p^4) := ((p^3)•p)                   exponent 4
    --> All the definitions are valid.
```

This is just a part of the extended operator set. Coefficients and exponents are defined recursively using plus and times. Three recurring steps (2, 3, and 4) are checked programmatically.

For coefficients, the actual recursive definitions are:

$$0p := \mathsf{F} \tag{3.3}$$

$$1p := p \tag{3.4}$$

$$kp := (k-1)p + p \qquad \forall(k > 1) \tag{3.5}$$

For exponents, the actual recursive definitions are:

$$p^0 := \mathsf{T} \tag{3.6}$$

$$p^1 := p \tag{3.7}$$

$$p^k := (k-1)p \times p \qquad \forall(k > 1) \tag{3.8}$$

The same definitions are used for each algebra which was tested.

3.2 Axioms

The tautologies for Ł$_\infty$ are also tautologies in the finite-valued Ł-algebras. In particular, we should verify (1) that the axioms for Ł$_\infty$ are satisfied, and also that *modus ponens* is a valid rule. These checks all passed as no error is displayed:

```
Axiom Schemas:
    A1: p⊃(q⊃p)                              positive paradox
    A2: (p⊃q)⊃((q⊃r)⊃(p⊃r))                    syllogism
    A3: (~p⊃~q)⊃(q⊃p)                        contraposition
    A4: (p∨q)⊃(q∨p)                          commutativity
    A5: ((p⊃q)⊃(q⊃p))⊃(q⊃p)
```

Note that A5 is equivalent by definition D2 to:

$$(p \supset q) \vee (q \supset p).$$

Also it is superfluous – it can be proved using A1 through A4 and *modus ponens*. We still need to check that *modus ponens* is satisfied:

```
Rules of Transformation:
    R1: If ⊢(p⊃q) and ⊢p, then ⊢q.                modus ponens
    --> All the transformations are valid.
```

So far, so good! Mordchaj Wajsberg's special axiom for Ł$_3$ was also satisfied (as expected):

```
    A6: ((p⊃~p)⊃p)⊃p                              Ł(3) axiom 4
```

Note that this is by definition equivalent to:

$$(p \supset \neg p) \vee p.$$

The next axioms to check are the coefficient axioms. The coefficient axiom for Ł$_2$ should fail, but the rest should all succeed. All of this set of axioms from Ł$_2$ through Ł$_{15}$ were checked. All except Ł$_2$ were satisfied:

```
    A7: (2p)⊃(1p)                                 Ł(2) ax6
        not valid: |p|=I, result=I
    A8: (3p)⊃(2p)                                 Ł(3) ax6
    A9: (4p)⊃(3p)                                 Ł(4) ax6
    A11: (5p)⊃(4p)                                Ł(5) ax6
    A13: (6p)⊃(5p)                                Ł(6) ax6
    A17: (7p)⊃(6p)                                Ł(7) ax6
       ... and so on ...
    A62: (14p)⊃(13p)                              Ł(14) ax6
    A74: (15p)⊃(14p)                              Ł(15) ax6
```

A7 is, by definition:

$$\text{A7.1}: \ (p + p) \supset p.$$

In Ł$_2$, this is materially equivalent to (but not equivalent by definition to):

$$\text{A7.2}: \ (p \vee p) \supset p.$$

A7.2 is valid in Ł$_3$ but A7.1 is not. A8, the coefficient axiom for Ł$_3$, is valid in both Ł$_2$ and Ł$_3$, but fails in all Ł-algebras which have four or more truth values.

We now need to check the exponent axioms that we found. There are none for either Ł$_2$ or Ł$_3$, but there are a little over 60 of these for Ł$_4$ through Ł$_1$5.

In Table 3.1, for each Ł$_n$, we list all values of j for which there is an exponent axiom. These are values of j which are strictly between 1 and $n-1$ and which are not divisors of $n-1$. Note that if n is odd, then $n-1$ is even – for odd values

n	j	failures
4	2	$j = 2$
5	3	None
6	2,3,4	$j = 2$
7	4,5	None
8	2,3,4,5,6	$j = 2$
9	3,5,6,7	None
10	2,4,5,6,7,8	$j = 2$
11	3,4,6,7,8,9	None
12	2,3,4,5,6,7,8,9,10	$j = 2$
13	5,7,8,9,10,11	None
14	2,3,4,5,6,7,8,9,10,11,12	$j = 2$
15	3,4,5,6,8,9,10,11,12,13	None

Table 3.1: Exponent axioms that fail in Ł$_3$

of n, j cannot be equal to 2. The $j = 2$ exponent axioms all fail in Ł$_3$ when n is between 3 and 16. The axioms for $j > 3$ are all satisfied in this interval. If this pattern hold for all $n > 3$, then for all $n > 3$, (1) when n is odd, Ł$_3$ has as tautologies all tautologies of Ł$_n$, and (2) when n is even, some tautologies of Ł$_n$ are not tautologies in Ł$_3$. We state this formally as a conjecture:

Conjecture 3.1 (Nesting for Ł$_3$)**.** *The tautologies of Ł$_n$ are all tautologies of Ł$_3$ if and only if either:*

- $n = \infty$*; or*
- $n \geqslant 3$ *and* $n \equiv 1 \pmod 2$.

When $n-1$ is prime, there is a full complement of axioms for values ranging from $j = 2$ through $j = n - 2$. Look at the second column in Table 3.1 when n is 4, 6, 8, 12, and 14.

Here are all the failures:

```
    A10: 3(((~p)^2)+(p·(1p)))                Ł(4) ax7/2
        not valid: |p|=I, result=F
    A14: 5(((~p)^2)+(p·(1p)))                Ł(6) ax7/2
        not valid: |p|=I, result=F
    A21: 7(((~p)^2)+(p·(1p)))                Ł(8) ax7/2
        not valid: |p|=I, result=F
    A32: 9(((~p)^2)+(p·(1p)))                Ł(10) ax7/2
        not valid: |p|=I, result=F
    A46: 11(((~p)^2)+(p·(1p)))               Ł(12) ax7/2
        not valid: |p|=I, result=F
    A63: 13(((~p)^2)+(p·(1p)))               Ł(14) ax7/2
        not valid: |p|=I, result=F
--> 7 axiom schemas are not valid.
```

In each case, the first failure detected was at $p = I$. A total of 84 axiom candidates were generated. Of these, 7 were not satisfied. One of these was a particular axiom of the coefficient type for Ł$_2$, the classical propositional calculus. The other 6 were exponent-type axioms for Ł$_n$ when n is even and larger than 3.

3.3 Theorems

About 55 tautologies of Ł$_2$ were checked against Ł$_3$. Seven of these failed. Here are the failures:

```
T5: (p⊃(q⊃r))⊃((p⊃q)⊃(p⊃r))              importation
    not valid: |r|=F, |q|=I, |p|=I, result=I
T9: ((p⊃q)⊃p)⊃p                          Pierce
    not valid: |p|=I, |q|=F, result=I
T11: (p⊃(q⊃r))⊃((p∧q)⊃r)                  importation
    not valid: |r|=F, |q|=I, |p|=I, result=I
T13: p∨~p                                excluded middle
    not valid: |p|=I, result=I
T14: ~(p∧~p)                             contradiction
    not valid: |p|=I, result=I
T15a: (p⊃~p)⊃~p                           reductio ad absurdum
    not valid: |p|=I, result=I
T15b: (~p⊃p)⊃p                           reductio ad absurdum
    not valid: |p|=I, result=I
```

Note that in T9 (Pierce's Law), if we substitute $\neg p$ for q, we get Wajsberg's particular axiom for Ł$_3$. (It appears above as Axiom A6.)

The two forms of *reductio ad absurdum* represent a basic contradiction argument in mathematics: Assume something is true and show that it must be false. Conclude then that it is indeed false. In Ł$_3$, if $|p| = I$, then $|p \supset \neg p| = T = |\neg p \supset p|$. So this argument form fails in Ł$_3$ (and in Ł$_n$ for all $n > 2$).

As a sidenote, T11 (imports) failed. But exports are valid:

```
T12: ((p∧q)⊃r)⊃(p⊃(q⊃r))                   exportation
```

In the Lewis modal system S2, for strict implication, imports are valid but imports are not:

$$S2 \vdash (p \dashv\exists (q \dashv\exists r)) \dashv\exists ((p \wedge q) \dashv\exists r) \qquad \text{imports (valid)}$$
$$((p \wedge q) \dashv\exists r) \dashv\exists (p \dashv\exists (q \dashv\exists r)) \qquad \text{exports (not valid)}$$

In Ł-algebras, negating something twice is the same in value as affirming it:

```
T17: p≡~~p                               double negation
```

(There are many-valued logics where this isn't valid, for example, the Gödel-Dummett algebras or the intuitionist propositional calculus.)[6]

3.4 Transformations

Eight rules of transformation were tested. These rules basically assert that if each of a set of wffs happens to be true under a particular assignment of the propositional variables, then the conclusion is true under the same assignment. One of the rules is *modus ponens* – for an Ł-algebra, this rule is valid. In addition, we check seven "derived rules" – these are additional rules of transformation which are classically valid, *i.e.* valid in $Ł_2$:

```
Rules of Transformation:
    R1: If ⊢(p⊃q) and ⊢p, then ⊢q.                  modus ponens
--> All the transformations are valid.
Derived Rules:
    DR1: If ⊢p and ⊢q, then ⊢(p∧q).                 adjunction
    DR2: If ⊢(p⊃q) and ⊢~q, then ⊢~p.               modus tollens
    DR3: If ⊢(~p⊃~q) and ⊢q, then ⊢p.               modus tollens
    DR4: If ⊢(p⊃~q) and ⊢q, then ⊢~p.               modus tollens
    DR5: If ⊢(~p⊃q) and ⊢~q, then ⊢p.               modus tollens
    DR6: If ⊢(~p⊃p), then ⊢p.                      contradiction
        Conclusion |p| = I when |p|=I and premises hold.
    DR7: If ⊢(p⊃~p), then ⊢~p.                     contradiction
        Conclusion |~p| = I when |p|=I and premises hold.
        --> 2 transformations are not valid.
```

In view of the fact that $\neg\neg p$ is equivalent to p in every Ł-algebra, The four forms of *modus tollens* are equivalent. These four rules are valid in all Ł-algebras since they follow from the contrapositive law.

The two contradiction rules are likewise equivalent. But neither is valid in

[6]The Gödel-Dummett algebras use different semantics for negation and implication. Negation has the sense of being impossible:

$$|\neg p| := \delta^1_{|p|} = \begin{cases} 1 \text{ if } |p| = 1; \text{ or} \\ 0 \text{ otherwise.} \end{cases}$$

Material implication differs in the handling of untrue cases:

$$|p \supset q| := \begin{cases} 1 \text{ if } |p| \geqslant |q|; \text{ or} \\ |q| \text{ otherwise.} \end{cases}$$

The truth values in the Gödel-Dummett algebra G_n are the same as those in $Ł_n$. The tautologies in the intuitionist propositional calculus form a *proper* subset of the tautologies in G_∞.

any Ł-algebra except $Ł_2$. Whenever $0 < |p| \leqslant 1/2$, we have:

$$1/2 \leqslant |\neg p| < 1;$$
$$|p \supset \neg p| = 1;$$

and thus:

$$\vdash\ p \supset \neg p.$$

But under the assumption, $\vdash \neg p$ is not true.

The adjunction rule is valid in all Ł-algebras since $T \wedge T$ is equivalent to T.

In our overall sample, sixteen $Ł_2$ tested facts are not valid in $Ł_3$.

4 Results for $Ł_4$

For $Ł_4$, the number of tested facts that are not valid is twenty.

4.1 Definitions

The tested definitions are all valid in $Ł_4$.

4.2 Axioms

Eleven of the axioms in the sample are not valid in $Ł_4$. (For $Ł_3$, seven were not valid.) The axioms for $Ł_\infty$ are, as expected, all valid. Wajsberg's $Ł_3$ axiom and the coefficient axiom for both $Ł_2$ and $Ł_3$ all fail, again as expected:

```
Axiom Schemas:
    A6: ((p⊃~p)⊃p)⊃p                        Wajsberg axion for Ł(3)
        not valid: |p|=2/3, result=2/3
    A7: (2p)⊃(1p)                           Ł(2) ax6
        not valid: |p|=1/3, result=2/3
    A8: (3p)⊃(2p)                           Ł(3) ax6
        not valid: |p|=1/3, result=2/3
```

The remaining coefficient axioms all pass, as expected. Now we look at the exponent axioms that fail. The results for the exponent axioms are summarized in Table 4.1.

In each of the exponent failures, the index was $j = 3$. The failures occurred whenever $4 - 1 = 3$ was not a divisor of $n - 1$. So the tautologies of $Ł_n$ are all tautologies of $Ł_4$ when n is 7, 10, or 13. If this pattern holds, then:

Conjecture 4.1 (Nesting for $Ł_4$)**.** *The tautologies of $Ł_n$ are all tautologies of $Ł_4$ if and only if either:*

n	j	failures
4	2	None
5	3	$j = 3$
6	2,3,4	$j = 3$
7	4,5	None
8	2,3,4,5,6	$j = 3$
9	3,5,6,7	$j = 3$
10	2,4,5,6,7,8	None
11	3,4,6,7,8,9	$j = 3$
12	2,3,4,5,6,7,8,9,10	$j = 3$
13	5,7,8,9,10,11	None
14	2,3,4,5,6,7,8,9,10,11,12	$j = 3$
15	3,4,5,6,8,9,10,11,12,13	$j = 3$

Table 4.1: Exponent axioms that fail in Ł$_4$

- $n = \infty$*; or*
- $n \geqslant 4$ *and* $n \equiv 1 \pmod 3$.

Here are the details:

```
A12: 4(((~p)^3)+(p·(2p)))                Ł(5) ax7/3
    not valid: |p|=1/3, result=0
A15: 5(((~p)^3)+(p·(2p)))                Ł(6) ax7/3
    not valid: |p|=1/3, result=0
A22: 7(((~p)^3)+(p·(2p)))                Ł(8) ax7/3
    not valid: |p|=1/3, result=0
A27: 8(((~p)^3)+(p·(2p)))                Ł(9) ax7/3
    not valid: |p|=1/3, result=0
A39: 10(((~p)^3)+(p·(2p)))               Ł(11) ax7/3
    not valid: |p|=1/3, result=0
A47: 11(((~p)^3)+(p·(2p)))               Ł(12) ax7/3
    not valid: |p|=1/3, result=0
A64: 13(((~p)^3)+(p·(2p)))               Ł(14) ax7/3
    not valid: |p|=1/3, result=0
A75: 14(((~p)^3)+(p·(2p)))               Ł(15) ax7/3
    not valid: |p|=1/3, result=0
```

4.3 Theorems

In our sample of Ł$_2$ tautologies, the same seven that failed in Ł$_3$ also failed in Ł$_4$. Of course the detected point of failure was different:

```
T5: (p⊃(q⊃r))⊃((p⊃q)⊃(p⊃r))          importation
```

```
    not valid: |p|=1/3, |r|=0, |q|=1/3, result=2/3
T9: ((p⊃q)⊃p)⊃p                          Pierce
    not valid: |p|=1/3, |q|=0, result=2/3
T11: (p⊃(q⊃r))⊃((p∧q)⊃r)                importation
    not valid: |q|=1/3, |p|=1/3, |r|=0, result=2/3
T13: p∨~p                                excluded middle
    not valid: |p|=1/3, result=2/3
T14: ~(p∧~p)                             contradiction
    not valid: |p|=1/3, result=2/3
T15: (p⊃~p)⊃~p                           reductio ad absurdum
    not valid: |p|=1/3, result=2/3
T15: (~p⊃p)⊃p                            reductio ad absurdum
    not valid: |p|=1/3, result=2/3
```

4.4 Transformations

As in Ł$_3$, the contradiction arguments both fail, but the *modus tollens* arguments and adjunction are all valid.

```
DR6: If ⊢(~p⊃p), then ⊢p.                                  contradiction
     Conclusion |p| = 2/3 when |p|=2/3 and premises hold.
DR7: If ⊢(p⊃~p), then ⊢~p.                                 contradiction
     Conclusion |~p| = 2/3 when |p|=1/3 and premises hold.
```

5 Results for Ł$_5$

The definitions are all valid in Ł$_5$. The same seven theorems and the same two derived rules fail. (The reader can work out the details of the failures.) We consider the axioms.

5.1 Axioms

Eighteen of the tested axioms are not valid. As expected, Wajsberg's axiom for Ł$_3$ and the coefficient axioms for Ł$_2$ through Ł$_4$ all fail:

```
A6: ((p⊃~p)⊃p)⊃p                          Ł(3) (Wajsberg's axiom)
    not valid: |p|=3/4, result=3/4
A7: (2p)⊃(1p)                            Ł(2) ax6
    not valid: |p|=1/4, result=3/4
A8: (3p)⊃(2p)                            Ł(3) ax6
    not valid: |p|=1/4, result=3/4
A9: (4p)⊃(3p)                            Ł(4) ax6
    not valid: |p|=1/4, result=3/4
```

n	j	failures
4	2	$j = 2$
5	3	None
6	2,3,4	$j \in \{2, 4\}$
7	4,5	$j = 4$
8	2,3,4,5,6	$j \in \{2, 4\}$
9	3,5,6,7	None
10	2,4,5,6,7,8	$j \in \{2, 4\}$
11	3,4,6,7,8,9	$j = 4$
12	2,3,4,5,6,7,8,9,10	$j \in \{2, 4\}$
13	5,7,8,9,10,11	None
14	2,3,4,5,6,7,8,9,10,11,12	$j \in \{2, 4\}$
15	3,4,5,6,8,9,10,11,12,13	$j = 4$

Table 5.1: Exponent axioms that fail in $Ł_5$

The results for the exponent axioms are summarized in Table 5.1.
The conjectured relationship is as follows:

Conjecture 5.1 (Nesting for $Ł_5$)**.** *The tautologies of $Ł_n$ are all tautologies of $Ł_5$ if and only if either:*

- $n = \infty$*; or*
- $n \geqslant 5$ *and* $n \equiv 1 \pmod 4$.

The details:

```
A10: 3(((~p)^2)+(p·(1p)))                Ł(4) ax7/2
     not valid: |p|=1/2, result=0
A14: 5(((~p)^2)+(p·(1p)))                Ł(6) ax7/2
     not valid: |p|=1/2, result=0
A16: 5(((~p)^4)+(p·(3p)))                Ł(6) ax7/4
     not valid: |p|=1/4, result=0
A18: 6(((~p)^4)+(p·(3p)))                Ł(7) ax7/4
     not valid: |p|=1/2, result=0
A21: 7(((~p)^2)+(p·(1p)))                Ł(8) ax7/2
     not valid: |p|=1/2, result=0
A23: 7(((~p)^4)+(p·(3p)))                Ł(8) ax7/4
     not valid: |p|=1/4, result=0
A32: 9(((~p)^2)+(p·(1p)))                Ł(10) ax7/2
     not valid: |p|=1/2, result=0
A33: 9(((~p)^4)+(p·(3p)))                Ł(10) ax7/4
     not valid: |p|=1/4, result=0
A40: 10(((~p)^4)+(p·(3p)))               Ł(11) ax7/4
     not valid: |p|=1/4, result=0
A46: 11(((~p)^2)+(p·(1p)))               Ł(12) ax7/2
```

```
     not valid: |p|=1/2, result=0
A48: 11(((~p)^4)+(p·(3p)))                Ł(12) ax7/4
     not valid: |p|=1/4, result=0
A63: 13(((~p)^2)+(p·(1p)))                Ł(14) ax7/2
     not valid: |p|=1/2, result=0
A65: 13(((~p)^4)+(p·(3p)))                Ł(14) ax7/4
     not valid: |p|=1/4, result=0
A76: 14(((~p)^4)+(p·(3p)))                Ł(15) ax7/4
     not valid: |p|=1/4, result=0
--> 18 axiom schemas are not valid.
```

6 Results for $Ł_6$

The definitions are all valid in $Ł_6$. The same seven theorems and the same two derived rules fail. (The reader can work out the details of the failures.) We consider the axioms.

6.1 Axioms

We have the expected failures for algebras with fewer truth values:

```
 A6: ((p⊃~p)⊃p)⊃p                         Wajsberg's Ł(3) axiom
     not valid: |p|=3/5, result=4/5
 A7: (2p)⊃(1p)                            Ł(2) ax6
     not valid: |p|=1/5, result=4/5
 A8: (3p)⊃(2p)                            Ł(3) ax6
     not valid: |p|=1/5, result=4/5
 A9: (4p)⊃(3p)                            Ł(4) ax6
     not valid: |p|=1/5, result=4/5
A11: (5p)⊃(4p)                            Ł(5) ax6
     not valid: |p|=1/5, result=4/5
```

Table 6.1 summarizes the results for the exponent axioms. From the table, we can make the following conjecture, consistent with the earlier conjectures:

Conjecture 6.1 (Nesting for $Ł_6$)**.** *The tautologies of $Ł_n$ are all tautologies of $Ł_6$ if and only if either:*

- $n = \infty$*; or*
- $n \geqslant 6$ *and* $n \equiv 1 \pmod 5$.

In addition, a pattern is starting to emerge in the values of the parameter j where the failures occur:

n	j	failures
4	2	$j = 2$
5	3	$j = 3$
6	2,3,4	None
7	4,5	$j = 5$
8	2,3,4,5,6	$j = 5$
9	3,5,6,7	$j = 5$
10	2,4,5,6,7,8	$j = 5$
11	3,4,6,7,8,9	None
12	2,3,4,5,6,7,8,9,10	$j = 5$
13	5,7,8,9,10,11	$j = 5$
14	2,3,4,5,6,7,8,9,10,11,12	$j = 5$
15	3,4,5,6,8,9,10,11,12,13	$j = 5$

Table 6.1: Exponent axioms that fail in $Ł_6$

index		failures
n	$n-1$	j
3	2	2
4	3	3
5	4	2, 4
6	5	5

Let's generalize this:

Conjecture 6.2 (Nesting)**.** *Let m and n be integers with $n \geqslant m \geqslant 3$. Then the tautologies of $Ł_n$ are all tautologies in $Ł_m$ if and only if $n \equiv 1 \pmod{m-1}$*

Here is our first conjecture for the failure nodes:

Conjecture 6.3 (Exponent axiom failures)**.** *Let m and n be integers with $n > m \geqslant 3$. Let j be an integer such that $1 < j < m-1$ and j does not divide $m-1$. Let X_j be the exponent axiom with index j in $Ł_n$. Then X_j is not a tautology in $Ł_m$ if and only if j divides $m-1$.*

Now for the details... First, there are failures of coefficient axioms in algebras with smaller index:

```
A10: 3(((~p)^2)+(p·(1p)))                Ł(4) ax7/2
     not valid: |p|=2/5, result=3/5
A12: 4(((~p)^3)+(p·(2p)))                Ł(5) ax7/3
     not valid: |p|=2/5, result=4/5
```

And there are failures in algebras with larger index:

```
   A19: 6(((~p)^5)+(p·(4p)))                Ł(7) ax7/5
        not valid: |p|=1/5, result=0
   A24: 7(((~p)^5)+(p·(4p)))                Ł(8) ax7/5
        not valid: |p|=1/5, result=0
   A28: 8(((~p)^5)+(p·(4p)))                Ł(9) ax7/5
        not valid: |p|=1/5, result=0
   A34: 9(((~p)^5)+(p·(4p)))                Ł(10) ax7/5
        not valid: |p|=1/5, result=0
   A49: 11(((~p)^5)+(p·(4p)))               Ł(12) ax7/5
        not valid: |p|=1/5, result=0
   A56: 12(((~p)^5)+(p·(4p)))               Ł(13) ax7/5
        not valid: |p|=1/5, result=0
   A66: 13(((~p)^5)+(p·(4p)))               Ł(14) ax7/5
        not valid: |p|=1/5, result=0
   A77: 14(((~p)^5)+(p·(4p)))               Ł(15) ax7/5
        not valid: |p|=1/5, result=0
--> 15 axiom schemas are not valid.
```

7 Results for Ł$_7$

We focus on the exponent axioms since there were no surprises in any of our other results for Ł$_7$. A total of twenty-seven sampled axioms were rejected. Twenty-one of these are exponent axioms. The results are summarized in Table 7.1.

n	j	failures
4	2	$j = 2$
5	3	$j = 3$
6	2,3,4	$j \in \{2, 3\}$
7	4,5	None
8	2,3,4,5,6	$j \in \{2, 3, 6\}$
9	3,5,6,7	$j \in \{3, 6\}$
10	2,4,5,6,7,8	$j \in \{2, 6\}$
11	3,4,6,7,8,9	$j \in \{3, 6\}$
12	2,3,4,5,6,7,8,9,10	$j \in \{2, 3, 6\}$
13	5,7,8,9,10,11	None
14	2,3,4,5,6,7,8,9,10,11,12	$j \in \{2, 3, 6\}$
15	3,4,5,6,8,9,10,11,12,13	$j \in \{3, 6\}$

Table 7.1: Exponent axioms that fail in Ł$_6$

At least for the sample, Conjectures 6.2 and 6.3 both continue to hold.

```
Axioms:
```

```
        --- Expected failures ---
    Wajsberg Ł(3): A6
    Coefficient: A7, A8, A9, A11, A13
--> Expected to fail: 6 axioms
        --- Lower index failures ---
   A10: 3(((~p)^2)+(p·(1p)))                Ł(4) ax7/2
        not valid: |p|=1/2, result=0
   A12: 4(((~p)^3)+(p·(2p)))                Ł(5) ax7/3
        not valid: |p|=1/3, result=0
   A14: 5(((~p)^2)+(p·(1p)))                Ł(6) ax7/2
        not valid: |p|=1/2, result=0
   A15: 5(((~p)^3)+(p·(2p)))                Ł(6) ax7/3
        not valid: |p|=1/3, result=0
        --- Higher index failures ---
   A21: 7(((~p)^2)+(p·(1p)))                Ł(8) ax7/2
        not valid: |p|=1/2, result=0
   A22: 7(((~p)^3)+(p·(2p)))                Ł(8) ax7/3
    not valid: |p|=1/3, result=0
   A25: 7(((~p)^6)+(p·(5p)))                Ł(8) ax7/6
        not valid: |p|=1/6, result=0
   A27: 8(((~p)^3)+(p·(2p)))                Ł(9) ax7/3
        not valid: |p|=1/3, result=0
   A29: 8(((~p)^6)+(p·(5p)))                Ł(9) ax7/6
        not valid: |p|=1/6, result=0
   A32: 9(((~p)^2)+(p·(1p)))                Ł(10) ax7/2
        not valid: |p|=1/2, result=0
   A35: 9(((~p)^6)+(p·(5p)))                Ł(10) ax7/6
        not valid: |p|=1/6, result=0
   A39: 10(((~p)^3)+(p·(2p)))               Ł(11) ax7/3
        not valid: |p|=1/3, result=0
   A41: 10(((~p)^6)+(p·(5p)))               Ł(11) ax7/6
        not valid: |p|=1/6, result=0
   A46: 11(((~p)^2)+(p·(1p)))               Ł(12) ax7/2
        not valid: |p|=1/2, result=0
   A47: 11(((~p)^3)+(p·(2p)))               Ł(12) ax7/3
        not valid: |p|=1/3, result=0
   A50: 11(((~p)^6)+(p·(5p)))               Ł(12) ax7/6
        not valid: |p|=1/6, result=0
   A63: 13(((~p)^2)+(p·(1p)))               Ł(14) ax7/2
        not valid: |p|=1/2, result=0
   A64: 13(((~p)^3)+(p·(2p)))               Ł(14) ax7/3
        not valid: |p|=1/6, result=0
   A67: 13(((~p)^6)+(p·(5p)))               Ł(14) ax7/6
        not valid: |p|=1/6, result=0
   A75: 14(((~p)^3)+(p·(2p)))               Ł(15) ax7/3
```

```
        not valid: |p|=1/3, result=0
   A78: 14(((~p)^6)+(p·(5p)))                Ł(15) ax7/6
        not valid: |p|=1/6, result=0
--> 27 axiom schemas are not valid.
```

8 Results for $Ł_8$

With two basic conjectures, namely Conjectures 6.2 on nesting and 6.3 on exponential axioms, we should see how they hold up in some additional cases. For $Ł_m$, when m is 8, 9, or 10, we expanded our checks up through $Ł_{20}$. The results for the sampled definitions, theorems, and rules of transformation were essentially unchanged – the definitions were all valid; the same theorems and rules failed, though the details of the failures (*i.e.* the truth values differed). So we again focus on the axioms. We will go through these for $Ł_8$ quickly, but in almost full detail.

As a review, we remark that the axioms for $Ł_\infty$ are all valid:

```
Axiom Schemas:
    A1: p⊃(q⊃p)                          positive paradox
    A2: (p⊃q)⊃((q⊃r)⊃(p⊃r))                syllogism
    A3: (~p⊃~q)⊃(q⊃p)                     contraposition
    A4: (p∨q)⊃(q∨p)                      commutativity
    A5: ((p⊃q)⊃(q⊃p))⊃(q⊃p)
```

Axiom A5 is equivalent by definition to:

$$(p \supset q) \vee (q \supset p),$$

and is dependent, *i.e.* A5 can be proved assuming A1-A4, *modus ponens*, and substitution.

Wajsberg's axiom, a special case of Pierce's Law[7] for $Ł_3$ fails:

```
    A6: ((p⊃~p)⊃p)⊃p                      Wajsberg's axiom Ł(3)
        not valid: |p|=4/7, result=6/7
```

The coefficient axioms for $Ł_n$ when n is less than 8 all fail. When $n \geqslant 8$, it succeeds:

[7] Pierce's Law:

$$((p \supset q) \supset p) \supset p$$

, or equivalently by definition

$$(p \supset q) \vee p$$

is valid in the classical 2-valued (true-false) case. If we substitute $\neq p$ for q, we have Wajsberg's axiom.

```
 A7: (2p)⊃(1p)                              Ł(2) ax6
     not valid: |p|=1/7, result=6/7
 A8: (3p)⊃(2p)                              Ł(3) ax6
     not valid: |p|=1/7, result=6/7
 A9: (4p)⊃(3p)                              Ł(4) ax6
     not valid: |p|=1/7, result=6/7
A11: (5p)⊃(4p)                              Ł(5) ax6
     not valid: |p|=1/7, result=6/7
A13: (6p)⊃(5p)                              Ł(6) ax6
     not valid: |p|=1/7, result=6/7
A17: (7p)⊃(6p)                              Ł(7) ax6
     not valid: |p|=1/7, result=6/7
A20: (8p)⊃(7p)                              Ł(8) ax6
A26: (9p)⊃(8p)                              Ł(9) ax6
A31: (10p)⊃(9p)                             Ł(10) ax6
        and so on through...
A131: (20p)⊃(19p)                           Ł(20) ax6
```

These results support the nesting conjecture, namely Conjecture 6.2. Finally we come to the exponent axioms. First we look at $Ł_n$ axioms when $n \leqslant 8$:

```
A10: 3(((~p)^2)+(p•(1p)))                   Ł(4) ax7/2
     not valid: |p|=3/7, result=3/7
A12: 4(((~p)^3)+(p•(2p)))                   Ł(5) ax7/3
     not valid: |p|=2/7, result=4/7
A14: 5(((~p)^2)+(p•(1p)))                   Ł(6) ax7/2
     not valid: |p|=3/7, result=5/7
A15: 5(((~p)^3)+(p•(2p)))                   Ł(6) ax7/3
     not valid: |p|=2/7, result=5/7
A16: 5(((~p)^4)+(p•(3p)))                   Ł(6) ax7/4
     not valid: |p|=2/7, result=5/7
A18: 6(((~p)^4)+(p•(3p)))                   Ł(7) ax7/4
     not valid: |p|=2/7, result=6/7
A19: 6(((~p)^5)+(p•(4p)))                   Ł(7) ax7/5 (valid)
A21: 7(((~p)^2)+(p•(1p)))                   Ł(8) ax7/2 (valid)
A22: 7(((~p)^3)+(p•(2p)))                   Ł(8) ax7/3 (valid)
A23: 7(((~p)^4)+(p•(3p)))                   Ł(8) ax7/4 (valid)
A24: 7(((~p)^5)+(p•(4p)))                   Ł(8) ax7/5 (valid)
A25: 7(((~p)^6)+(p•(5p)))                   Ł(8) ax7/6 (valid)
```

Note that 8-1=7 which is prime – for $Ł_8$, there are five exponent axioms, one for each $j \in \{2, 3, 4, 5, 6\}$. We summarize the exponent axiom failures in Table 8.1.

From Conjecture 6.3, we expect to see failures for $n > 8$ when n is not congruent to 1 modulo 7. In the table, we expect to see no failures when $n = 15$. Since 7 is prime, the only failure when $n > 8$ should be when $j = 7$. Table 8.1 supports Conjecture 6.3. Here is a record of the failures for $n > 8$:

n	j	failures
4	2	$j = 2$
5	3	$j = 3$
6	2,3,4	$j \in \{2, 3, 4\}$
7	4,5	$j = 4$
8	2,3,4,5,6	None
9	3,5,6,7	$j = 7$
10	2,4,5,6,7,8	$j = 7$
11	3,4,6,7,8,9	$j = 7$
12	2,3,4,5,6,7,8,9,10	$j = 7$
13	5,7,8,9,10,11	$j = 7$
14	2,3,4,5,6,7,8,9,10,11,12	$j = 7$
15	3,4,5,6,8,9,10,11,12,13	None
16	2,4,6,7,8,9,10,11,12,13,14	$j = 7$
17	3,5,6,7,9,10,11,12,13,14,15	$j = 7$
18	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16	$j = 7$
19	4,5,7,8,10,11,12,13,14,15,16,17	$j = 7$
20	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18	$j = 7$

Table 8.1: Exponent axioms that fail in $Ł_8$

```
A30: 8(((~p)^7)+(p·(6p)))                Ł(9) ax7/7
    not valid: |p|=1/7, result=0
A36: 9(((~p)^7)+(p·(6p)))                Ł(10) ax7/7
    not valid: |p|=1/7, result=0
A42: 10(((~p)^7)+(p·(6p)))               Ł(11) ax7/7
    not valid: |p|=1/7, result=0
A51: 11(((~p)^7)+(p·(6p)))               Ł(12) ax7/7
    not valid: |p|=1/7, result=0
A57: 12(((~p)^7)+(p·(6p)))               Ł(13) ax7/7
     not valid: |p|=1/7, result=0
A68: 13(((~p)^7)+(p·(6p)))               Ł(14) ax7/7
    not valid: |p|=1/7, result=0
```

For $Ł_{15}$, each of the axioms passes. (There is no axiom for $j = 7$ since 7 divides 15-1=14.)

```
 A89: 15(((~p)^7)+(p·(6p)))              Ł(16) ax7/7
     not valid: |p|=1/7, result=0
A101: 16(((~p)^7)+(p·(6p)))              Ł(17) ax7/7
     not valid: |p|=1/7, result=0
A115: 17(((~p)^7)+(p·(6p)))              Ł(18) ax7/7
     not valid: |p|=1/7, result=0
A128: 18(((~p)^7)+(p·(6p)))              Ł(19) ax7/7
```

```
      not valid: |p|=1/7, result=0
A134: 19(((~p)^7)+(p·(6p)))                Ł(20) ax7/7
      not valid: |p|=1/7, result=0
--> 24 axiom schemas are not valid.
```

In all, 155 axioms were sampled, 24 of which were not valid in $Ł_8$.

9 Results for $Ł_9$ and $Ł_{10}$

For $Ł_9$ and $Ł_{10}$ we note, respectively, that 42 and 38 of the 155 sampled axioms were not valid. Both Conjecture 6.2 and 6.3 hold within this sampling. Tables 9.1 and 9.2 summarize the sample results for the exponent axioms. For $n > m > 9$, we expect $Ł_{17}$ to nest inside $Ł_9$ since $17 \equiv 1 \pmod 8$ and failures should occur at $j \in \{2, 4, 8\}$. For $n > m > 10$, we expect $Ł_{19}$ to nest inside $Ł_9$ since $19 \equiv 1 \pmod 9$ and failures should occur at $j \in \{3, 9\}$.

n	j	failures ($m = 9$)
4	2	$j = 2$
5	3	$j = 3$
6	2,3,4	$j \in \{2, 3, 4\}$
7	4,5	$j = 4$
8	2,3,4,5,6	$j \in \{2, 3, 4\}$
9	3,5,6,7	None
10	2,4,5,6,7,8	$j \in \{2, 4, 8\}$
11	3,4,6,7,8,9	$j \in \{4, 8\}$
12	2,3,4,5,6,7,8,9,10	$j \in \{2, 4, 8\}$
13	5,7,8,9,10,11	$j = 8$
14	2,3,4,5,6,7,8,9,10,11,12	$j \in \{2, 4, 8\}$
15	3,4,5,6,8,9,10,11,12,13	$j \in \{4, 8\}$
16	2,4,6,7,8,9,10,11,12,13,14	$j \in \{2, 4, 8\}$
17	3,5,6,7,9,10,11,12,13,14,15	None
18	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16	$j \in \{2, 4, 8\}$
19	4,5,7,8,10,11,12,13,14,15,16,17	$j \in \{4, 8\}$
20	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18	$j \in \{2, 4, 8\}$

Table 9.1: Exponent axioms that fail in $Ł_9$

$\mathfrak{n}$	$\mathfrak{j}$	failures ($\mathfrak{m} = 10$)
4	2	$\mathfrak{j} = 2$
5	3	$\mathfrak{j} = 3$
6	2,3,4	$\mathfrak{j} \in \{2, 3, 4\}$
7	4,5	$\mathfrak{j} \in \{4, 5\}$
8	2,3,4,5,6	$\mathfrak{j} \in \{2, 3, 4, 5\}$
9	3,5,6,7	$\mathfrak{j} \in \{3, 5\}$
10	2,4,5,6,7,8	None
11	3,4,6,7,8,9	$\mathfrak{j} \in \{3, 9\}$
12	2,3,4,5,6,7,8,9,10	$\mathfrak{j} \in \{3, 9\}$
13	5,7,8,9,10,11	$\mathfrak{j} = 9$
14	2,3,4,5,6,7,8,9,10,11,12	$\mathfrak{j} \in \{3, 9\}$
15	3,4,5,6,8,9,10,11,12,13	$\mathfrak{j} \in \{3, 9\}$
16	2,4,6,7,8,9,10,11,12,13,14	$\mathfrak{j} = 9$
17	3,5,6,7,9,10,11,12,13,14,15	$\mathfrak{j} \in \{3, 9\}$
18	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16	$\mathfrak{j} \in \{3, 9\}$
19	4,5,7,8,10,11,12,13,14,15,16,17	None
20	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18	$\mathfrak{j} \in \{3, 9\}$

Table 9.2: Exponent axioms that fail in $Ł_1 0$

10 Divisor lattices and Ł-algebras

10.1 Divisor lattices

Let n be any positive integer greater than 1. Let D_n be the set of positive integers which are divisors of n. Let i and j be two integers in D_n. The join of i and j, denoted $i \vee j$, is the least common multiple of both i and j. The meet of i and j, denoted $i \wedge j$, is the greatest common divisor.

For example, let $n = 12 = 2^2 \times 3$.

$$D_{12} = \{1, 2, 3, 4, 6, 12\}.$$

We can map this set using what's known as a Hasse diagram. The Hasse diagram for D_{12} is mapped in Figure 10.1. For numbers a and b in the diagram, a divides b if and only if there is a rising path from a to b. (Rising paths include trivial paths – the rising paths that start at 2 end at 2, 4, 6 and 12, but never at 1 or 3.)

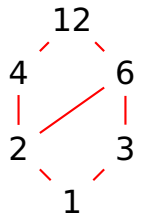


Figure 10.1: Hasse diagram for the divisor lattice D_{12}

The complement (or codivisor) of $i \in D_{12}$ is defined as $12/i$ and can be denoted as $\neg i$.[8] Table 10.1 displays matrices for the join, meet, and complement operations for D_12.

The standard way to define an *order* on a lattice is to use either the join or the meet operator:

$$i \leqslant j \text{ iff } i \vee j = j \text{; or. equivalently:} \tag{10.1}$$

$$i \leqslant j \text{ iff } i \wedge j = i. \tag{10.2}$$

For a divisor lattice, the first definition says that the least common multiple of i and j is j, while the second says the greatest common divisor of i and j is i. Either one amounts to saying that i divides j. To turn this divisor algebra into a many-valued logic, we need to define a material implication operator which is compatible with the order operator. By "compatible", I mean that the result is true whenever the operands are in order and has a value other than true when

[8] The name "codivisor" is short for complementary divisor. Note that the set of codivisors is the same as the set of divisors.

$i \vee j$	j 1	2	3	4	6	12
i 1	1	2	3	4	6	12
2	2	2	6	4	6	12
3	3	6	3	12	6	12
4	4	4	12	4	12	12
6	6	6	6	12	6	12
12	12	12	12	12	12	12

$i \wedge j$	j 1	2	3	4	6	12
i 1	1	1	1	1	1	1
2	1	2	1	2	2	2
3	1	1	3	1	3	3
4	1	2	1	4	2	1
6	1	2	3	2	6	6
12	1	2	3	4	6	12

i	1	2	3	4	6	12
$\neg i$	12	6	4	3	2	1

Table 10.1: Lattice operations for D_{12}

the operands are not in order. (There are many ways to do this, but we have a special one in mind.)

In particular, we want the the implication operator $\supset$ to satisfy the following relation:

$$i \vee j = (i \supset j) \supset j.$$

Based on our definitions so far, our true state is 12 and our false state is 1.

$i \supset j$	j 1	2	3	4	6	12
i 1	12	12	12	12	12	12
2	?	12	?	12	12	12
3	?	?	12	?	12	12
4	?	?	?	12	?	12
6	?	?	?	?	12	12
12	?	?	?	?	?	12

$i \vee j$	j 1	2	3	4	6	12
i 1	1	2	3	4	6	12
2	2	2	6	4	6	12
3	3	6	3	12	6	12
4	4	4	12	4	12	12
6	6	6	6	12	6	12
12	12	12	12	12	12	12

Table 10.2: Material implication in D_{12}

Comparing the two matrices in Table 10.2, to get the first row in the second matrix we need:

$$12 \supset i = i.$$

This fills the last row of the matrix on the left. In addition, we can fill the first column by assuming that contrapositives are equal in value:

$$i \supset 1 = 12/i = \neg i.$$

These two adjustments leave us with nine unassigned entries in the implication matrix, none of which can be equal to 12. Although we can manually reduce the

number of possibilities, it's easier to reduce the number of possibilities using a computer program. The output is as follows:[9]

```
contrapositive C12,2=2=C6,1
contrapositive C12,3=3=C4,1
contrapositive C12,4=4=C3,1
contrapositive C12,6=6=C2,1
4 fixes...

     (p⊃q)            (p∨q) := ((p⊃q)⊃q)   NOTE 1
12 12 12 12 12 12       1  2  3  4  6 12
 6 12  ? 12 12 12       2  2  ?  4  6 12
 4  ? 12  ? 12 12       3  ?  3  ?  6 12
 3  ?  ? 12  ? 12       4  ?  ?  4  ? 12
 2  ?  ?  ? 12 12       6  ?  ?  ?  6 12
 1  2  3  4  6 12      12 12 12 12 12 12

9 unassigned values
1953125 functions                            NOTE 2
Pass 1: 15625 functions    (~p⊃~q)≡(q⊃p)     NOTE 3
Pass 2: 6 functions     (p∨q) := ((p⊃q)⊃q)   NOTE 4
Pass 3: 1 functions        p⊃(q⊃p)           NOTE 5

   (p⊃q)  index=1858098
12 12 12 12 12 12
 6 12  6 12 12 12
 4  4 12  4 12 12
 3  6  3 12  6 12
 2  4  6  4 12 12
 1  2  3  4  6 12
```

Notes on the display:

1. Displayed here are the matrices with the last row and first column fixed as described. We use the definition and the known values to obtain as much of the disjunction matrix as possible. There are nine entries in the matrix on the left that need to be determined.
2. In principle, each of the undetermined entries could be any non-true value. There are five values: $\{1, 2, 3, 4, 6\}$. There are 1,953,125 ways of mapping the undetermined entries into this set of values.
3. In pass 1, we check that the assigned entries satisfy the contrapositive law. (This is an tautology in every Ł-algebra.) At the end of pass 1, we are left with 15,625 possible mappings,
4. In pass 2, we check whether the assignments produce the correct values for disjunction, namely the least common multiples. (This is pass 2 since

[9] See Python module `logic.D12imp`.

this check takes more computer processing time than the check in pass 1.) At the end of this check we have six remaining candidates.

5. In pass 3, we check that the positive paradox is satisfied. We are left with just one possibility for material implication.

We label the rows and columns of the implication matrix in Table 10.3. In addition, we write the entries a products of primes.

$i \supset j$	1	2	3	4	6	12
1	12	12	12	12	12	12
2	6	12	6	12	12	12
3	4	4	12	4	12	12
4	3	6	3	12	6	12
6	2	4	6	4	12	12
12	1	2	3	4	6	12

$i \supset j$	1	2	3	2^2	$2 \cdot 3$	$2^2 \cdot 3$
$i = 1$	$2^2 \cdot 3$	$2^2 \cdot 3$	$2^2 \cdot 3$	$2^2 \cdot 3$	$2^2 \cdot 3$	$2^2 \cdot 3$
2	$2 \cdot 3$	$2^2 \cdot 3$	$2 \cdot 3$	$2^2 \cdot 3$	$2^2 \cdot 3$	$2^2 \cdot 3$
3	2^2	2^2	$2^2 \cdot 3$	2^2	$2^2 \cdot 3$	$2^2 \cdot 3$
2^2	3	$2 \cdot 3$	3	$2^2 \cdot 3$	$2 \cdot 3$	$2^2 \cdot 3$
$2 \cdot 3$	2	2^2	$2 \cdot 3$	2^2	$2^2 \cdot 3$	$2^2 \cdot 3$
$2^2 \cdot 3$	1	2	3	2^2	$2 \cdot 3$	$2^2 \cdot 3$

Table 10.3: Material implication in D_{12}

In fact, we separate the matrices for implication and negation in powers of 2 and powers of 3. This division logic is a product of two logics. For powers of two, we write F=0, I=2 and T=4, while for powers of 3 we write F=0 and T=3. The separated matrices are in Table 10.4.

power of 2 logic

$i \supset j$	F	I	T
F	T	T	T
I	I	T	T
T	F	I	T

p	$\neg p$
F	T
I	I
T	F

power of 3 logic

$i \supset j$	F	T
F	T	T
T	F	T

p	$\neg p$
F	T
T	T

Table 10.4: D_{12} as a product of Ł$_3$ and Ł$_2$

The divisor logic D_12 is just a product of two Ł-algebras. The "power of two logic" under a relabelling is just Ł$_3$ and the power of three logic is just the classical propositional calculus (or Ł$_2$).

10.2 Finite Ł-algebras as divisor lattices

Let p be any prime positive integer and let n be any integer larger than 1. Consider the Ł-algebra $Ł_{\mathrm{n}}$. We can define a partial order on the truth values for $Ł_{\mathrm{n}}$ by:

$$|\mathrm{p}| \leqslant |\mathrm{q}| \quad \text{iff} \quad \vdash |\mathrm{p} \supset \mathrm{q}| = \mathrm{T}) \tag{10.3}$$

The resulting structure is a divisor lattice on the truth values, The join and meet operations are $|\mathrm{p} \vee \mathrm{q}|$ and $|\mathrm{p} \wedge \mathrm{q}|$. The lattice is isomorphic to $\mathrm{D}_{\mathrm{p}^{\mathrm{n}-1}}$.

This explains why the nesting conjecture (Conjecture 6.2) is valid.[10] Of course this analysis doesn't help with the with the exponent conjecture.

10.3 Divisor logics and modal logic

We turned the divisor lattice D_{12} into a divisor logic by defining negation and material implication so that they satisfied certain properties. The properties were chosen so that are resulting system satisfied properties valid in $Ł_{\infty}$. Further analysis revealed that the resulting algebra was really a product of a logic isomorphic to $Ł_3$ and another which was isomorphic to $Ł_2$.

Now, suppose we think of one of these logics as representing the actual state of affairs (say $Ł_3$) and the other logic (*i.e.* $Ł_2$) as a state of affairs in a parallel universe. We have six truth states:

$$(\mathrm{F}, \mathrm{F}),\ (\mathrm{F}, \mathrm{T}),\ (\mathrm{I}, \mathrm{F}),\ (\mathrm{I}, \mathrm{T}),\ (\mathrm{T}, \mathrm{F}),\ \text{and}\ (\mathrm{T}, \mathrm{T}).$$

The rules for manipulating the first state are perfectly encapsulated in $Ł_3$, while the second state follows the usual rule for the classical propositional calculus (or $Ł_2$.) We can create functionally independent operators which describe concepts like actually true (T in this universe regardless of the state in the other universe), necessarily true (T in both universes), and possibly true (T in at least one universe). This sort of approach was used in Lewis and Langford to model the semantics[11] of the Lewis modal systems S1 through S5.[2] The various worlds used classical true-false logic. Our logic based on D_{12} is a two-universe logic which uses two different Ł-algebras, one isomorphic to the clasical propositional calculus, the other isomorphic to the Ł-algebra $Ł_3$.

We are not limited to Ł-algebras. Any algebra which is compatible with a prime-power divisor lattice could be used – the definitions of disjunction (as lcm) and conjunction (as gcd) are fixed but there is considerable flexibility in defining implication and negation.

A Python source

The Python modules may be found on *Github* in the following repository:

[10] In other words, Conjecture 6.2 is a theorem.

[11] For example: to show that certain theorems one system were not valid in another system.

- https://github.com/econrad003/science

(The boxed text above in blue is a clickable link. It should take you to the *Github* repository. The blue text below is not clickable.)

The Python modules used here are found in the logic folder – these are listed here using Python module syntax:

- logic.lukasiewicz – basic setup for a many-world Ł-algebra
- logic.grigolia1977 – adds what is required to evaluate the operators use to develop Grigolia's 1977 axioms for $Ł_m$
- logic.D12imp – script to reduce roughly two million possibilities down to one (see Section 10)
- logic.tests.luk2 – script to produce the data used in the "Results" sections; the actual results have been edited for inclusion here

The actual results produced by the logic.tests.luk2 script are found in the messages folder:

- test_luk2.txt – results for $Ł_3$ through $Ł_7$
- test_luk2b.txt – results for $Ł_8$ through $Ł_{10}$

References

[1] Herbert Hendry. “Functional completeness and non-Łukasiewiczian truth functions” in the *Notre Dame Journal of Formal Logic*, **v21** n3 (July 1980). Offprint.

[2] Clarence Lewis and Cooper Langford. *Symbolic logic*, Second Edition. Dover (NY, 1959).

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